

232.2.4971 - Quantum computation

Homework Assignment 01

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Question 01 - Inner Products, Angles and Measurement of a single qubit

Let $|\psi\rangle = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$.

- Write the state $|\psi\rangle$ using the standard notation. Write $\langle\psi|$ using the standard notation.
- Verify that $\| |\psi\rangle \|^2 \equiv \langle\psi|\psi\rangle = 1$.
- What is the inner product between $|\psi\rangle$ and $|0\rangle$? What is the angle between $|\psi\rangle$ and $|0\rangle$? (Note: the angle between two vectors u, v is defined as $\arccos\left(\frac{\langle u, v \rangle}{\|u\| \cdot \|v\|}\right)$)
- You are given a quantum state which is either $|\psi\rangle$ or $|0\rangle$ uniformly at random. You win if you are able to identify correctly the state that you are given, and you lose otherwise. Calculate the winning probabilities of the following strategies:
 - Measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer '1' if the outcome is '1'.
 - Apply the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, and then measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer '1' if the outcome is '1'.

Answer

- in standard notation $|\psi\rangle$ is $\begin{bmatrix} \cos(\pi/8) \\ \sin(\pi/8) \end{bmatrix}$, and $\langle\psi|$ is $\begin{bmatrix} \cos(\pi/8) & \sin(\pi/8) \end{bmatrix}$.

- Let's verify $\| |\psi\rangle \|^2 = 1$:

$$\| |\psi\rangle \|^2 = \langle\psi|\psi\rangle = \begin{bmatrix} \cos(\pi/8) & \sin(\pi/8) \end{bmatrix} \begin{bmatrix} \cos(\pi/8) \\ \sin(\pi/8) \end{bmatrix} = \cos^2(\pi/8) + \sin^2(\pi/8) = 1$$

- The inner product between $|\psi\rangle$ and $|0\rangle$ is:

$$\begin{aligned} \langle\psi|0\rangle &= \cos(\pi/8) \langle 0|0\rangle + \sin(\pi/8) \langle 0|1\rangle \\ &= \cos(\pi/8) \end{aligned}$$

The angle is:

$$\begin{aligned} \frac{\langle\psi|0\rangle}{\| |\psi\rangle \| \| |0\rangle \|} &= \frac{\cos(\pi/8)}{|\langle\psi|\psi\rangle|^2 \cdot |\langle 0|0\rangle|^2} \\ &= \frac{\cos(\pi/8)}{|1|^2 \cdot |1|^2} \\ &= \cos(\pi/8) \end{aligned}$$

- (a) Let Π_0 and Π_1 be the projection operator for the computational basis, then the probability of winning is:

$$\begin{aligned} \Pr\{win\} &= \Pr\{win | |0\rangle\} \Pr\{|0\rangle\} + \Pr\{win | |\psi\rangle\} \Pr\{|\psi\rangle\} \\ &= \langle 0|\Pi_0|0\rangle / 2 + \langle\psi|\Pi_1|\psi\rangle / 2 \\ &= |\langle 0|0\rangle|^2 / 2 + |\langle 1|\psi\rangle|^2 / 2 \\ &= (1 + \sin^2(\pi/8)) / 2 \approx 0.57 \end{aligned}$$

- (b) Let Π_0 and Π_1 be the projection operator for the computational basis, then the probability of winning is:

$$\begin{aligned}
\Pr\{win\} &= \Pr\{win \mid |0\rangle\} \Pr\{|0\rangle\} + \Pr\{win \mid |\psi\rangle\} \Pr\{|\psi\rangle\} \\
&= \langle 0 | H \Pi_1 H | 0 \rangle / 2 + \langle \psi | H \Pi_0 H | \psi \rangle / 2 \\
&= (\langle 0 | + \langle 1 |) | 1 \rangle \langle 1 | (| 0 \rangle + | 1 \rangle) / 4 \\
&\quad + ((\cos(\pi/8) + \sin(\pi/8)) \langle 0 | + (\cos(\pi/8) - \sin(\pi/8)) \langle 1 |) \\
&\quad | 0 \rangle \langle 0 | \\
&\quad (\cos(\pi/8) + \sin(\pi/8)) | 0 \rangle + (\cos(\pi/8) - \sin(\pi/8)) | 1 \rangle) / 4 \\
&= |\langle 1 | 1 \rangle|^2 / 4 + |(\cos(\pi/8) + \sin(\pi/8)) \langle 0 | 0 \rangle|^2 / 4 \\
&= (1 + 1 + 2 \cos(\pi/8) \sin(\pi/8)) / 4 \\
&= 1/2 + \sin(\pi/4) / 4 \approx 0.68
\end{aligned}$$

We can see that this strategy is better than the previous one.

Question 2 - Tensor Product State

Let $|\alpha\rangle = a|0\rangle + b|1\rangle$, $|\beta\rangle = c|0\rangle + d|1\rangle$, where $a, b, c, d \in \mathbb{C}$ (Notice that this vector may not be normalized).

- Write the state $|\alpha\beta\rangle$ in the $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ basis, using both the Dirac notation and the usual matrix notation.
- Calculate the standard inner product between $|\alpha\rangle$ and $|\beta\rangle$.
- Calculate the norm of $|\alpha\rangle$ and prove that $\| |\alpha\beta\rangle \|^2 = \| |\alpha\rangle \|^2 \cdot \| |\beta\rangle \|^2$.
- Prove or disprove: Is the state $|\psi\rangle = \sqrt{\frac{1}{3}}|01\rangle + \sqrt{\frac{2}{3}}|10\rangle$ a tensor product state?
- Same question as previous item, but for the state $|\psi\rangle = \sqrt{\frac{1}{3}}|11\rangle + \sqrt{\frac{2}{3}}|10\rangle$.

Answer

- a) In Dirac notation:

$$\begin{aligned}
|\alpha\beta\rangle &= (a|0\rangle + b|1\rangle)(c|0\rangle + d|1\rangle) \\
&= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle
\end{aligned}$$

In matrix notation:

$$\begin{aligned}
|\alpha\beta\rangle &= \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} \\
&= \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}
\end{aligned}$$

- b) The inner product is:

$$\begin{aligned}
\langle \alpha | \beta \rangle &= (a^* \quad b^*) \begin{pmatrix} c \\ d \end{pmatrix} \\
&= a^*c + b^*d
\end{aligned}$$

- c) The norm of $|\alpha\rangle$ is:

$$\begin{aligned}
\| |\alpha\rangle \| &= \sqrt{\langle \alpha | \alpha \rangle} \\
&= \sqrt{(a^* \quad b^*) \begin{pmatrix} a \\ b \end{pmatrix}} \\
&= \sqrt{|a|^2 + |b|^2}
\end{aligned}$$

prove that $\|\alpha\beta\|^2 = \|\alpha\|^2 \cdot \|\beta\|^2$:

$$\begin{aligned}\|\alpha\beta\|^2 &= \langle\alpha\beta|\alpha\beta\rangle \\ &= \langle\alpha|\alpha\rangle \langle\beta|\beta\rangle \\ &= \|\alpha\|^2 \cdot \|\beta\|^2\end{aligned}$$

d) This is not a tensor product state:

A tensor product state is of the form $|\alpha\beta\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$, therefore for $|\psi\rangle$ to be a tensor product state the following must hold:

$$\begin{aligned}ac &= 0 \\ ad &= \sqrt{\frac{1}{3}} \\ bc &= \sqrt{\frac{2}{3}} \\ bd &= 0\end{aligned}$$

There is no solution for this set of equations since from $ac = 0$ either a or c is 0 but this contradicts the fact that both ad and bc are not 0 item[e)] This is a tensor product state:

A tensor product state is of the form $|\alpha\beta\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$, therefore for $|\psi\rangle$ to be a tensor product state the following must hold:

$$\begin{aligned}ac &= 0 \\ ad &= 0 \\ bc &= \sqrt{\frac{2}{3}} \\ bd &= \sqrt{\frac{1}{3}}\end{aligned}$$

a possible solution for this set of equations is: $\begin{matrix} a=0 \\ b=1 \\ c=\sqrt{\frac{2}{3}} \\ d=\sqrt{\frac{1}{3}} \end{matrix}$ and therefore this state is a tensor product state.

Question 3 - Operations and Measurements of Multiple Qubits

Let $|\psi\rangle = \sqrt{\frac{1}{6}}(|00\rangle - 2|10\rangle + |11\rangle)$

- Write the state $|\psi\rangle$ in the standard notation (i.e., as a column vector).
- We measure both the qubits of the state $|\psi\rangle$ in the $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ basis. What are the various probabilities of the result?
- We measure only the right qubit of the state $|\psi\rangle$. What are the probabilities of the outcomes 0 and 1? What is the state of the two qubits after the measurement, given that we measured “0”? Write the result also in the standard notation.
- What happens to the state if we first apply a Hadamard gate on the left qubit and only then measure the right qubit? Answer the same questions as in section (c).

Answer

a)

$$|\psi\rangle = \sqrt{\frac{1}{6}} \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

b) Let Π_e be the projection operator where $e \in |00\rangle, |01\rangle, |10\rangle, |11\rangle$, then:

$$\begin{aligned}\Pr(00) &= \langle \psi | \Pi_{00} | \psi \rangle = \langle 00 | \psi \rangle^2 = \left(\sqrt{\frac{1}{6}} \right)^2 = \frac{1}{6} \\ \Pr(01) &= \langle \psi | \Pi_{01} | \psi \rangle = \langle 01 | \psi \rangle^2 = 0^2 = 0 \\ \Pr(10) &= \langle \psi | \Pi_{10} | \psi \rangle = \langle 10 | \psi \rangle^2 = \left(-2\sqrt{\frac{1}{6}} \right)^2 = \frac{4}{6} \\ \Pr(11) &= \langle \psi | \Pi_{11} | \psi \rangle = \langle 11 | \psi \rangle^2 = \left(\sqrt{\frac{1}{6}} \right)^2 = \frac{1}{6}\end{aligned}$$

c) Let Π_e be the projection operator where $e \in |0\rangle, |1\rangle$ and I the single qubit identity operator, then the probabilities are:

$$\begin{aligned}\Pr(00 \vee 10) &= \langle \psi | I \otimes \Pi_0 | \psi \rangle = \langle 00 | \psi \rangle^2 + \langle 10 | \psi \rangle^2 = \left(\sqrt{\frac{1}{6}} \right)^2 + \left(-2\sqrt{\frac{1}{6}} \right)^2 = \frac{5}{6} \\ \Pr(01 \vee 11) &= \langle \psi | I \otimes \Pi_1 | \psi \rangle = \langle 01 | \psi \rangle^2 + \langle 11 | \psi \rangle^2 = 0^2 + \left(\sqrt{\frac{1}{6}} \right)^2 = \frac{1}{6}\end{aligned}$$

The new state is:

$$I \otimes \Pi_0 | \psi \rangle = \sqrt{\frac{1}{6}} (|00\rangle - 2|10\rangle)$$

And after normalization:

$$\begin{aligned}\frac{\sqrt{\frac{1}{6}} (|00\rangle - 2|10\rangle)}{\sqrt{\frac{5}{6}}} &= \sqrt{\frac{1}{5}} (|00\rangle - 2|10\rangle) \\ &= \sqrt{\frac{1}{5}} \begin{pmatrix} 1 \\ 0 \\ -2 \\ 0 \end{pmatrix}\end{aligned}$$

d) Let Π_e be the projection operator where $e \in |0\rangle, |1\rangle$, I the single qubit identity operator and H the Hadamard operator, then:

$$\begin{aligned}|\phi\rangle &= H \otimes I | \psi \rangle \\ &= \sqrt{\frac{1}{6}} \frac{1}{\sqrt{2}} ((|0\rangle + |1\rangle) \otimes |0\rangle - 2(|0\rangle - |1\rangle) \otimes |0\rangle + (|0\rangle - |1\rangle) \otimes |1\rangle) \\ &= \sqrt{\frac{1}{12}} (-|00\rangle + 3|10\rangle + |01\rangle - |11\rangle)\end{aligned}$$

Measuring the new state, the probabilities are:

$$\begin{aligned}\Pr(00 \vee 10) &= \langle \phi | I \otimes \Pi_0 | \phi \rangle = \langle 00 | \phi \rangle^2 + \langle 10 | \phi \rangle^2 = \left(-\sqrt{\frac{1}{12}} \right)^2 + \left(3\sqrt{\frac{1}{12}} \right)^2 = \frac{10}{12} \\ \Pr(01 \vee 11) &= \langle \phi | I \otimes \Pi_1 | \phi \rangle = \langle 01 | \phi \rangle^2 + \langle 11 | \phi \rangle^2 = \left(\sqrt{\frac{1}{12}} \right)^2 + \left(-\sqrt{\frac{1}{12}} \right)^2 = \frac{2}{12}\end{aligned}$$

The new state is:

$$I \otimes \Pi_0 |\phi\rangle = \sqrt{\frac{1}{12}} (-|00\rangle + 3|10\rangle)$$

And after normalization:

$$\begin{aligned} \frac{\sqrt{\frac{1}{6}}(-|00\rangle + 3|10\rangle)}{\sqrt{\frac{10}{12}}} &= \sqrt{\frac{1}{10}}(-|00\rangle + 3|10\rangle) \\ &= \sqrt{\frac{1}{10}} \begin{pmatrix} -1 \\ 0 \\ 3 \\ 0 \end{pmatrix} \end{aligned}$$

Question 4 - Transformations of Bell's basis

In the lecture, we defined four states:

$$\begin{aligned} |\psi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ |\psi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\phi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |\phi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned}$$

These states are often called Bell's basis. Suppose Alice holds the left qubit of $|\psi^+\rangle$ and Bob holds the other qubit.

- Show how Alice can transform $|\psi^+\rangle$ to the three other states, by acting only on her qubit.
- We defined the goal of super-dense coding in the last lecture before passover. Explain how to do super-dense coding. In more detail, Alice and Bob share the $|\psi^+\rangle$ state. Alice has a two-bit message $m = (m_1, m_2)$. Alice is allowed to use their entangled state and send one qubit to Bob. Bob needs to recover m . Design such a protocol.

Answer

- The transformation to the other states are as follows:

$$\begin{aligned} |\psi^+\rangle &\rightarrow |\psi^+\rangle : I \otimes I |\psi^+\rangle \\ |\psi^+\rangle &\rightarrow |\psi^-\rangle : Z \otimes I |\psi^+\rangle \\ |\psi^+\rangle &\rightarrow |\phi^+\rangle : X \otimes I |\psi^+\rangle \\ |\psi^+\rangle &\rightarrow |\phi^-\rangle : ZX \otimes I |\psi^+\rangle \end{aligned}$$

Where X, Z are (in the computational basis):

$$\begin{aligned} X &= \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \\ Z &= \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \end{aligned}$$

- The protocol is as follows:

- 0) Assume Alice and Bob share the state $|\psi^+\rangle$. This can be done by Alice performing the Hadamard operator on the state $|00\rangle$ and then sending the second qubit to Bob.
- 1) Alice wants to send Bob a two bits message (which corresponds to one of the four Bell's basis). She performs a transformation from the original state to the new state using the operators described in the previous section (those operators act only on her qubit and leave Bob's unchanged).
- 2) Alice sends her altered qubit to Bob which now holds both qubits (i.e. the entire state). Bob then performs $CNOT$ and then Hadamard operators which return the qubit from Bell's basis state to the computational state.
- 3) Bob measures the state and receives the two bit information.